Example 1.17

The product of two rational numbers is $\frac{-2}{3}$. If one number is $\frac{3}{7}$, then find the other.

Solution:

Let the other number be x.

Given, $\frac{3}{7} \times x = \frac{-2}{3}$

Multiplying by the reciprocal of $\frac{3}{7}$, that is $\frac{7}{3}$ on both sides,

$$\Rightarrow \frac{7}{3} \times \frac{3}{7} \times x = \frac{7}{3} \times \frac{-2}{3}$$

$$\Rightarrow x = \frac{-14}{9}$$

Aliter

$\frac{3}{7} \times x = \frac{-2}{3}$

$\Rightarrow x = \frac{-2}{3} \times \frac{7}{3} = \frac{-14}{9}$

(by cross multiplication)

Example 1.18

One roll of ribbon is $18\frac{3}{4}$ m long. Sankari has four full rolls and one–third of another roll. How many metres of ribbon does Sankari have in total?



Fig. 1.15

Solution:

Number of metres of ribbon Sankari has in total

$$= 18\frac{3}{4} \times 4\frac{1}{3}$$

$$= \frac{75}{4} \times \frac{13}{3} = \frac{325}{4} = 81\frac{1}{4} \text{ m}$$

Example 1.19

Find the rational numbers that should be added and subtracted so that they will make the sum $3\frac{1}{2} + 1\frac{3}{4} + 2\frac{3}{8}$ to the nearest whole number.

Solution:

Now, $3\frac{1}{2} + 1\frac{3}{4} + 2\frac{3}{8}$

$$= \frac{7}{2} + \frac{7}{4} + \frac{19}{8} = \frac{7 \times 4 + 7 \times 2 + 19 \times 1}{8} = \frac{28 + 14 + 19}{8}$$

$= \frac{61}{8} = 7\frac{5}{8}$, which lies between the whole numbers 7 and 8.

If we subtract $\frac{5}{8}$ from $7\frac{5}{8}$, it becomes 7. If we add $\frac{3}{8}$ to $7\frac{5}{8}$, it becomes $7+\frac{5}{8}+\frac{3}{8}=7+1=8$.

Example 1.20

A student instead of multiplying a number by $\frac{8}{9}$, by mistake divided it by $\frac{8}{9}$. If the difference between the correct answer and the answer got by him is 34, then find the number.

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